Student Name: Muhammad Hassaan Ahmad

Registration / Roll No: FA24-BCS-070

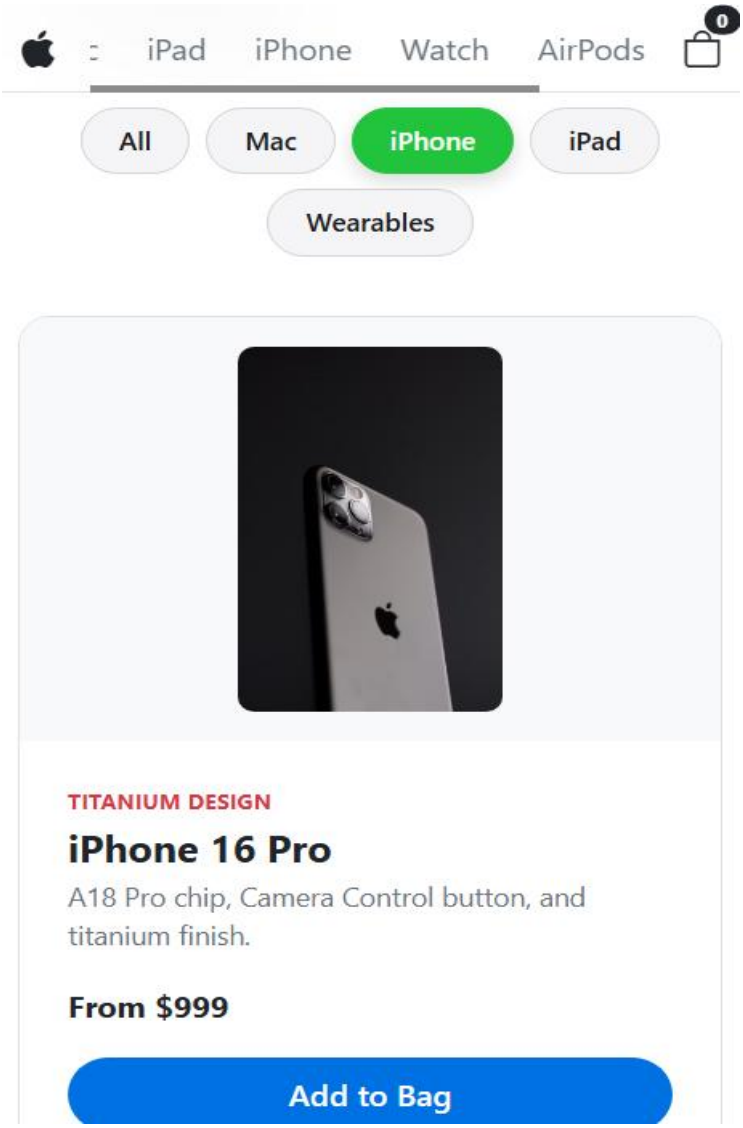
Course: CSC336 - Web Technologies

Exam: FA26 BCS A Lab Mid Term

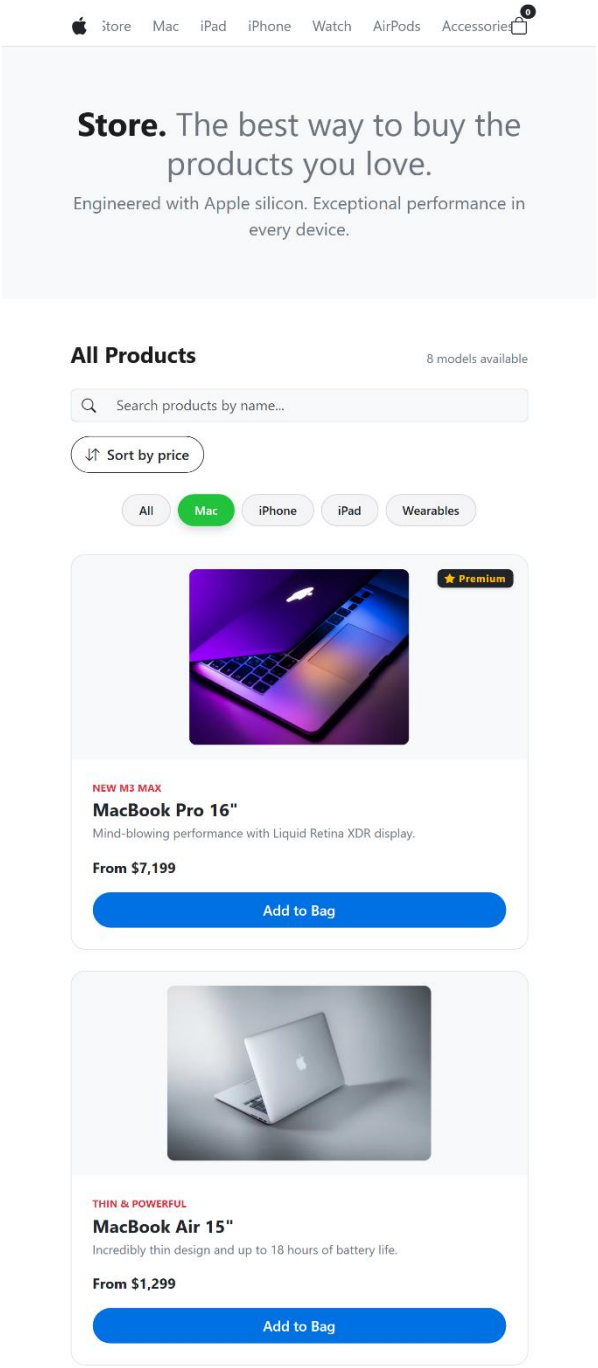
Personal Constants:

- $N = 70$ (Last 2 digits)
- Breakpoint $BP = 600 + N = 670$px
- Accent Hue $H = (70 \times 7) \mod 360 = 130° \Longrightarrow$ hsl$(130, 70\%, 45\%)$
- Categories $K = (70 \mod 3) + 3 = 4$
- Task 2 Sticky Offset = $(70 \mod 40) + 70$px $= 100$px
- Task 3 Premium Price Threshold = $N \times 100 = \$7{,}000$

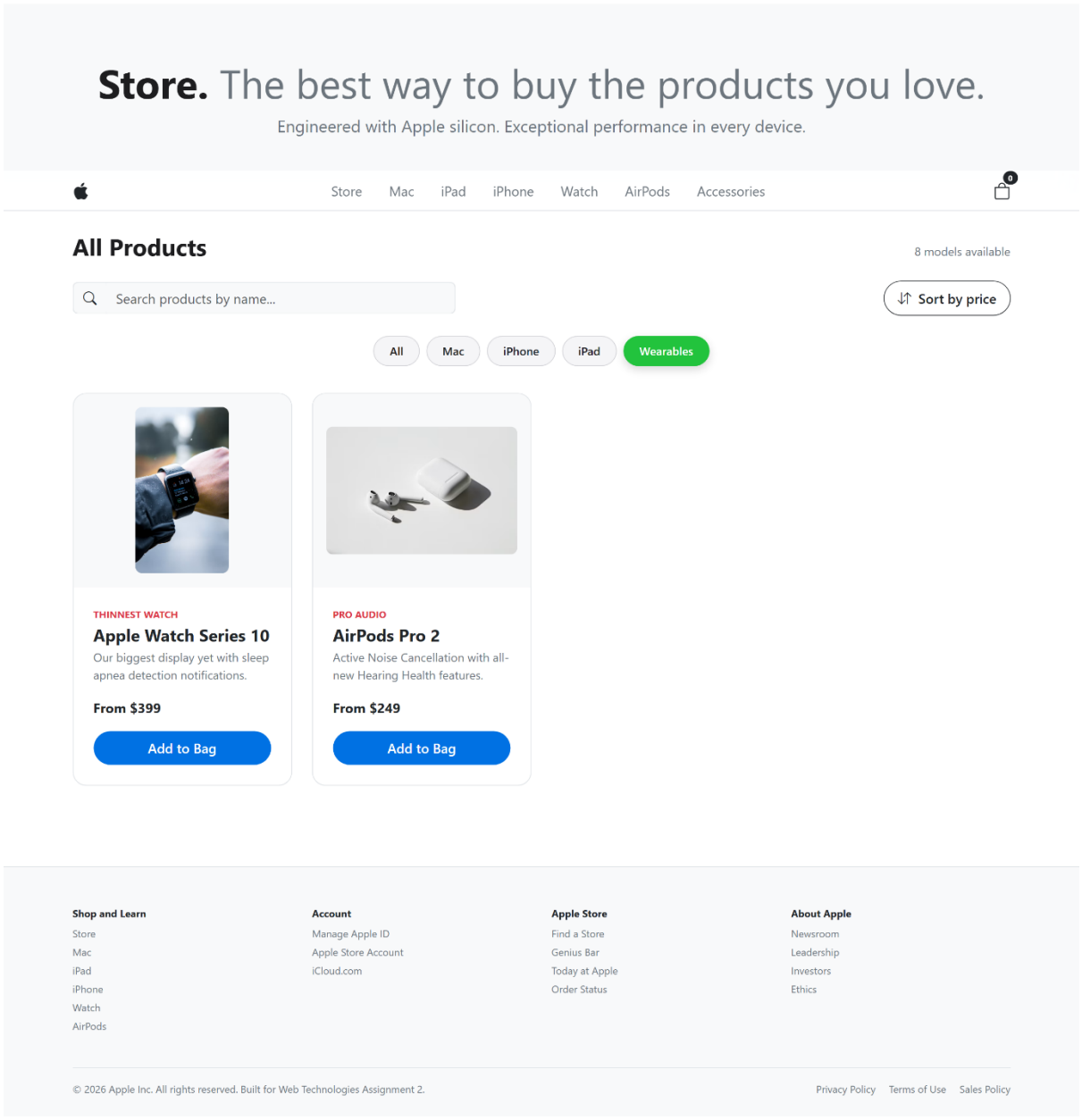
TASK 1 SCREENSHOTS:


iPad
iPhone
Watch
AirPods
0
All
Mac
iPhone
iPad
Wearables
TITANIUM DESIGN
iPhone 16 Pro
A18 Pro chip, Camera Control button, and titanium finish.
From $999
Add to Bag

375px


Store Mac iPad iPhone Watch AirPods Accessories 0
Store. The best way to buy the products you love.
Engineered with Apple silicon. Exceptional performance in every device.
All Products
8 models available
Search products by name...
Sort by price
All
Mac
iPhone
iPad
Wearables
Premium
NEW M3 MAX
MacBook Pro 16"
Mind-blowing performance with Liquid Retina XDR display.
From $7,199
Add to Bag
THIN & POWERFUL
MacBook Air 15"
Incredibly thin design and up to 18 hours of battery life.
From $1,299
Add to Bag

670px



1280px

TASK2 SCREENSHOTS:

iPad iPhone Watch AirPods

Store / Mac / MacBook Pro 16"



NEW M3 MAX SILICON

MacBook Pro 16"

Space Black finish with Liquid Retina XDR display

$7,199.00

Finish

Space Black | Silver

Unified Memory

128GB Unified Memory — Included

Quantity

1 — Free delivery & returns

Add to Bag

Delivers in 2–3 business days with free setup.

Product Overview & Architecture

The 16-inch MacBook Pro blasts forward with M3 Max, an extraordinarily advanced chip that brings massive performance and capabilities for extreme workflows. Built on industry-leading 3-nanometer technology and featuring next-generation GPU architecture with hardware-accelerated ray tracing and mesh shading, it renders complex 3D scenes up to 2.5x faster than previous generation silicon.

Enjoy extreme dynamic range with the Liquid Retina XDR display, featuring 1,000 nits sustained full-screen brightness for HDR content, up to 1,600 nits peak brightness, and 10,000 mini-LEDs grouped into individually controlled local dimming zones. With ProMotion technology offering adaptive refresh rates up to 120Hz, scrolling and gaming feel extraordinarily fluid and responsive.

Connect everything you need with three Thunderbolt 4 ports, an HDMI port that supports 8K displays at up to 60Hz or 4K displays at 240Hz, an SDXC card slot for fast photo and video transfer, a 3.5mm headphone jack with advanced support for high-impedance headphones, and a dedicated MagSafe 3 charging port with fast-charge capability.

Technical Specifications

Processor
Apple M3 Max chip with 16-core CPU, 40-core GPU, and 16-core Neural Engine

Display
16.2-inch Liquid Retina XDR; 3456-by-2234 native resolution; 1,000,000:1 contrast ratio

Memory & Storage
128GB unified memory with 400GB/s bandwidth; 2TB ultrafast SSD

Battery & Power
Up to 22 hours Apple TV playback; 100-watt-hour lithium-polymer battery

Audio & Camera
Studio-quality three-mic array; Six-speaker sound system with force-cancelling woofers

Wireless
Wi-Fi 6E (802.11ax) and Bluetooth 5.3

375px



NEW M3 MAX SILICON

MacBook Pro 16"

Space Black finish with Liquid Retina XDR display

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670px



NEW M3 MAX SILICON

MacBook Pro 16"

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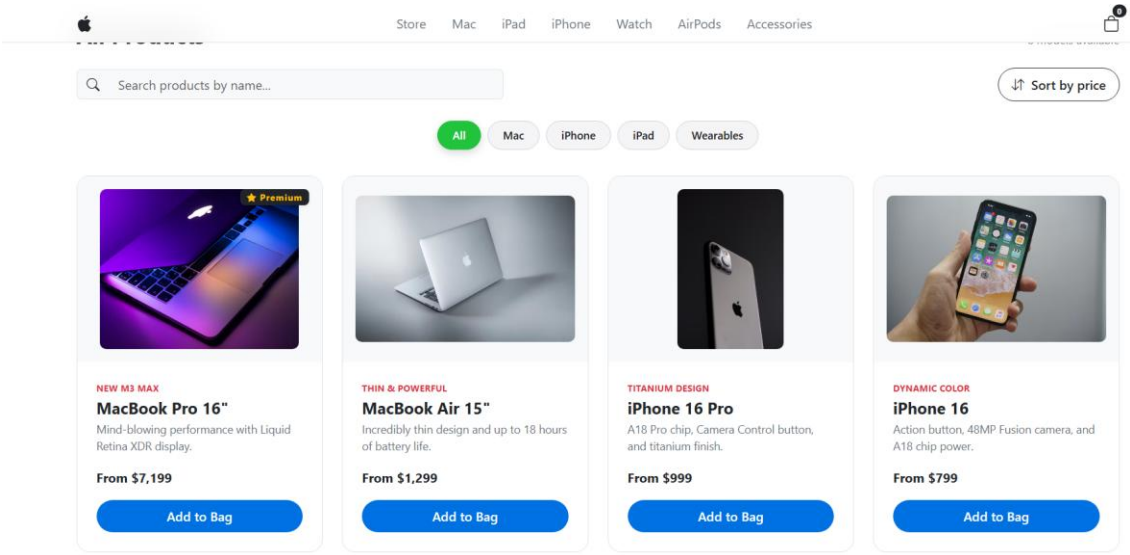
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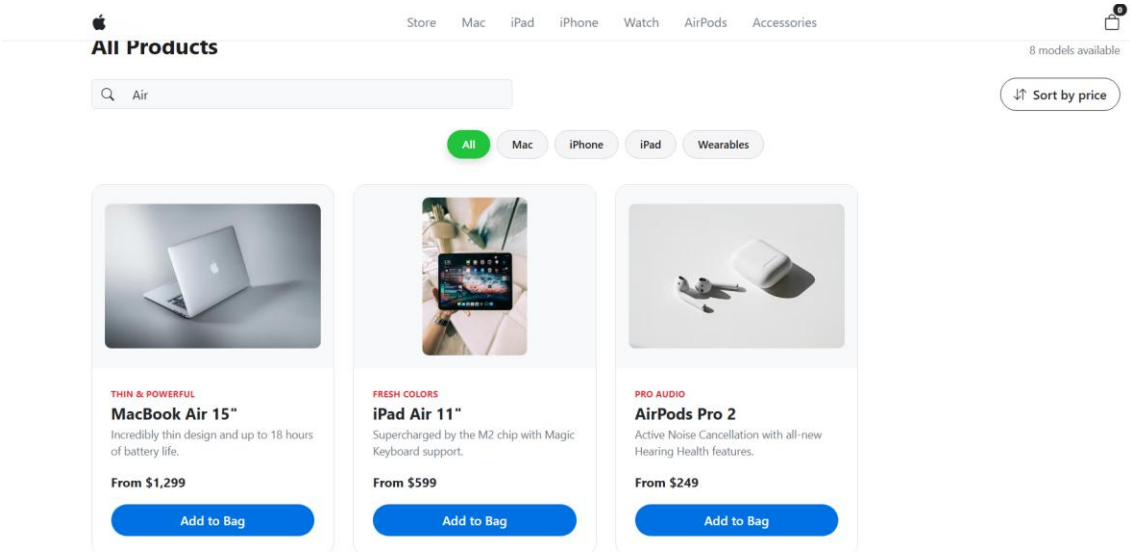
© 2026 Apple Inc. All rights reserved. Built for Web Technologies Assignment 2.

1280px

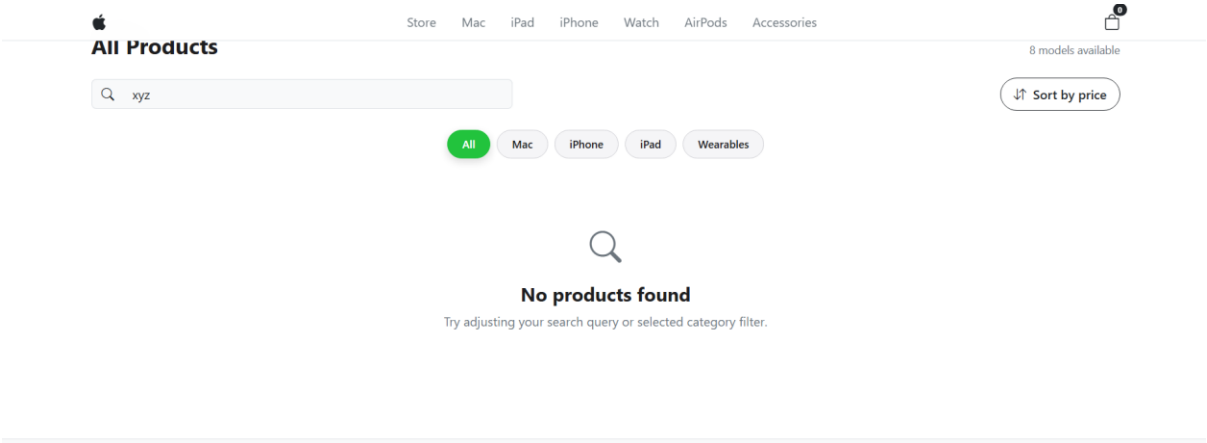
TASK 3 SCREENSHOTS:



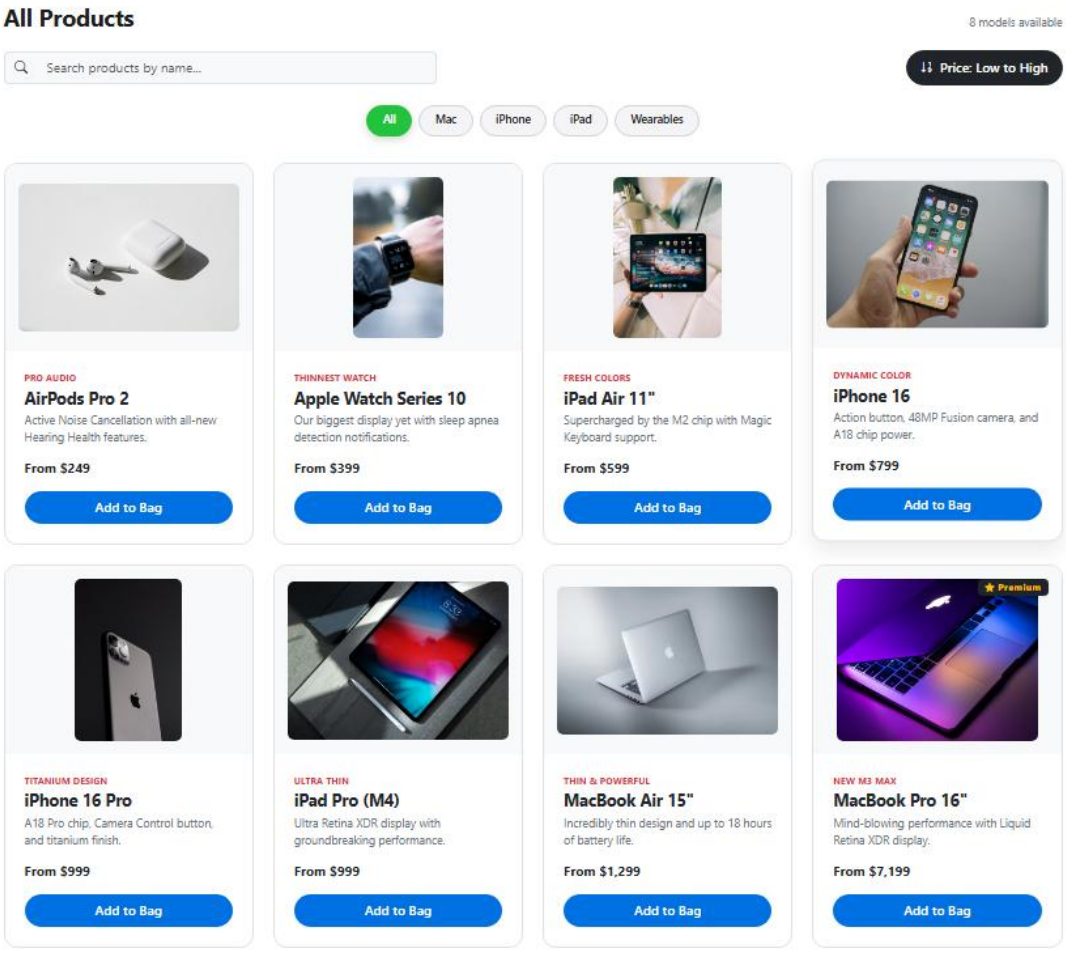
Premium Product Marked



Live Search



No Product Found Confirmation



Sort By Price

CODE MODIFICATIONS:

Task 1 — CSS-Only Category Filter:

index.html:

```
<!-- Category Filter Radio Buttons (K = 4 Categories + All) -->
<div id="filter-container" class="category-filter-wrapper">
  <input type="radio" name="category" id="cat-all" value="all" checked>
  <label for="cat-all" class="filter-btn">All</label>

  <input type="radio" name="category" id="cat-mac" value="mac">
  <label for="cat-mac" class="filter-btn">Mac</label>

  <input type="radio" name="category" id="cat-iphone" value="iphone">
  <label for="cat-iphone" class="filter-btn">iPhone</label>

  <input type="radio" name="category" id="cat-ipad" value="ipad">
  <label for="cat-ipad" class="filter-btn">iPad</label>

  <input type="radio" name="category" id="cat-wearables" value="wearables">
  <label for="cat-wearables" class="filter-btn">Wearables</label>
</div>

<!-- Product Cards Updated with Category & Price Attributes -->
<div class="col product-col" data-category="mac" data-price="7199">...</div>
<div class="col product-col" data-category="mac" data-price="1299">...</div>
<div class="col product-col" data-category="iphone" data-price="999">...</div>
<div class="col product-col" data-category="iphone" data-price="799">...</div>
<div class="col product-col" data-category="ipad" data-price="999">...</div>
```

```
<div class="col product-col" data-category="ipad" data-price="599">...</div>
<div class="col product-col" data-category="wearables" data-price="399">...</div>
<div class="col product-col" data-category="wearables" data-price="249">...</div>
```

CSS Rules Added (css/styles.css):

```
/* CSS-Only Category Filter (Accent Hue H = 130: hsl(130, 70%, 45%)) */
:root {
  --accent-color: hsl(130, 70%, 45%);
}

.category-filter-wrapper {
  display: flex;
  flex-wrap: wrap;
  gap: 0.5rem;
  justify-content: center;
  margin-bottom: 2rem;
}

.category-filter-wrapper input[type="radio"] {
  position: absolute;
  opacity: 0;
  pointer-events: none;
}

.filter-btn {
  display: inline-block;
  padding: 0.45rem 1.15rem;
  font-size: 0.85rem;
```

```
  font-weight: 500;
  color: #1d1d1f;
  background-color: #f5f5f7;
  border: 1px solid #d2d2d7;
  border-radius: 980px;
  cursor: pointer;
  transition: all 0.2s ease;
}

.filter-btn:hover {
  background-color: #e8e8ed;
}

/* Selected state with accent color */
.category-filter-wrapper input[type="radio"]:checked + .filter-btn {
  background-color: var(--accent-color) !important;
  border-color: var(--accent-color) !important;
  color: #ffffff !important;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.15);
}

/* Zero-JS Filtering via :has() and General Sibling Combinator (~) */
#filter-container:has(#cat-mac:checked) ~ #productGrid .product-col:not([data-category="mac"]) {
  display: none !important;
}

#filter-container:has(#cat-iphone:checked) ~ #productGrid .product-col:not([data-category="iphone"]) {
```

```
  display: none !important;
}

#filter-container:has(#cat-ipad:checked) ~ #productGrid .product-col:not([data-category="ipad"]) {
  display: none !important;
}

#filter-container:has(#cat-wearables:checked) ~ #productGrid .product-col:not([data-category="wearables"]) {
  display: none !important;
}
```

Task 2 — CSS Grid Layout with Sticky Details Box:

product.html:

```
<main class="py-4 py-lg-5">
  <div class="container">
    <div class="product-grid-layout">

      <!-- Area 1: Image Showcase (grid-area: image) -->
      <div class="area-image">
        <div class="bg-body-tertiary rounded-4 p-4 p-lg-5 d-flex align-items-center justify-content-center">
          <img src="https://images.unsplash.com/photo-1517336714731-489689fd1ca8?auto=format&fit=crop&w=1200&q=80" class="img-fluid rounded-3 object-fit-contain mh-100" alt="MacBook Pro 16-inch">
        </div>
      </div>
```

```
    <!-- Area 2: Sticky Details Box (grid-area: details) -->
    <div class="area-details bg-white p-3 p-md-4 rounded-4 border shadow-sm">
      <span class="badge text-bg-warning text-uppercase px-2 py-1 mb-2">New M3 Max Silicon</span>
      <h1 class="display-6 fw-bold mb-1">MacBook Pro 16"</h1>
      <p class="text-secondary mb-2">Space Black finish with Liquid Retina XDR display</p>
      <div class="fs-2 fw-bold text-dark mb-3">$7,199.00</div>

      <!-- Finish, Memory, and Quantity Form -->
      <form class="mb-3" onsubmit="event.preventDefault();">
        <div class="mb-3">
          <label for="productQuantity" class="form-label fw-semibold">Quantity</label>
          <div class="d-flex align-items-center gap-3">
            <input type="number" id="productQuantity" name="quantity" class="form-control text-center fw-bold" value="1" min="1" max="10" style="max-width: 90px;">
            <span class="text-secondary small">Free delivery &amp;amp; returns</span>
          </div>
        </div>
        <div class="d-grid gap-2">
          <button type="button" class="btn btn-apple w-100 py-2 fw-medium border-0">Add to Bag</button>
        </div>
      </form>
    </div>

    <!-- Area 3: Full-Width Description (grid-area: description) -->
    <div class="area-description border-top pt-4 mt-2">
```

```
      <h2 class="h3 fw-bold mb-3">Product Overview &amp;amp; Architecture</h2>
      <p class="text-secondary fs-6 leading-relaxed mb-4">
       The 16-inch MacBook Pro blasts forward with M3 Max, an extraordinarily advanced chip...
      </p>
      <!-- Extended paragraphs and technical specifications dl list -->
     </div>

   </div>
  </div>
</main>
```

CSS Rules Added (css/styles.css):

```
/* Product Detail Page CSS Grid Layout */
.product-grid-layout {
  display: grid;
  gap: 2rem;
  /* Below BP (670px): stacked 1 column */
  grid-template-areas:
    "image"
    "details"
    "description";
  grid-template-columns: 1fr;
}

.area-image {
  grid-area: image;
}
```

```
.area-details {
  grid-area: details;
}

.area-description {
  grid-area: description;
}

/* At Breakpoint BP = 670px and wider */
@media (min-width: 670px) {
  .product-grid-layout {
    grid-template-columns: 1.1fr 0.9fr;
    grid-template-areas:
      "image details"
      "description description";
    gap: 2.5rem;
    align-items: start;
  }

  /* Details box sticky at (N mod 40) + 70 = 100px */
  .area-details {
    position: sticky;
    top: 100px;
    z-index: 10;
    align-self: start;
  }
}
```

Task 3 — JavaScript: Live Search, Price Sort & Premium Badges:

index.html:

```
<!-- Search Input & Sort Button Controls -->
<div class="row g-3 justify-content-between align-items-center mb-4">
  <div class="col-12 col-md-5">
    <div class="input-group">
      <span class="input-group-text bg-light border-end-0"><i class="bi bi-search"></i></span>
      <input type="text" id="productSearch" class="form-control bg-light border-start-0" placeholder="Search products by name...">
    </div>
  </div>

  <div class="col-12 col-md-3 text-md-end">
    <button type="button" id="sortPriceBtn" class="btn btn-outline-dark rounded-pill px-3 py-2 small fw-semibold">
      <i class="bi bi-arrow-down-up me-1"></i> Sort by price
    </button>
  </div>
</div>

<!-- "No products found" placeholder -->
<div id="noProductsMsg" class="d-none text-center py-5">
  <i class="bi bi-search fs-1 text-secondary mb-3 d-block"></i>
  <h4 class="fw-bold">No products found</h4>
  <p class="text-secondary">Try adjusting your search query or selected category filter.</p>
```

```
</div>

<!-- Script Inclusion right before </body> -->
<script src="script.js"></script>
```

script.js:

```
/* Task 3: Live Search, Price Sorting & Premium Labels (N = 70, Threshold = $7,000) */
document.addEventListener("DOMContentLoaded", () => {
  const N = 70;
  const PREMIUM_THRESHOLD = N * 100; // $7,000

  const searchInput = document.getElementById("productSearch");
  const sortBtn = document.getElementById("sortPriceBtn");
  const productGrid = document.getElementById("productGrid");
  const noProductsMsg = document.getElementById("noProductsMsg");
  const categoryRadios = document.querySelectorAll('input[name="category"]');
  const productCards = Array.from(document.querySelectorAll(".product-col"));

  let sortDirection = "none";

  // 1. Automatic "Premium" Label on Page Load for Price > $7,000
  productCards.forEach((col) => {
    const price = parseFloat(col.dataset.price) || 0;
    if (price > PREMIUM_THRESHOLD) {
      const card = col.querySelector(".product-card");
      if (card && !card.querySelector(".premium-badge")) {
        const badge = document.createElement("span");
```

```
      badge.className = "badge bg-dark text-warning position-absolute top-0 end-0 m-3 px-2 py-1 fw-bold premium-badge shadow-sm";
      badge.style.fontSize = "0.72rem";
      badge.style.letterSpacing = "0.05em";
      badge.style.zIndex = "5";
      badge.innerHTML = '<i class="bi bi-star-fill me-1 text-warning"></i>Premium';
      card.appendChild(badge);
    }
  }
 });

 // 2. Synchronized Live Search + Category Filter
 function filterProducts() {
  const query = searchInput.value.trim().toLowerCase();
  const activeRadio = document.querySelector('input[name="category"]:checked');
  const selectedCategory = activeRadio ? activeRadio.value : "all";

  let visibleCount = 0;

  productCards.forEach((col) => {
   const titleElem = col.querySelector(".product-title");
   const title = titleElem ? titleElem.textContent.toLowerCase() : "";
   const category = col.dataset.category;

   const matchesSearch = title.includes(query);
   const matchesCategory = selectedCategory === "all" || category === selectedCategory;

   if (matchesSearch && matchesCategory) {
```

```
      col.style.setProperty("display", "block", "important");
      visibleCount++;
    } else {
      col.style.setProperty("display", "none", "important");
    }
  });

  if (visibleCount === 0) {
    noProductsMsg.classList.remove("d-none");
  } else {
    noProductsMsg.classList.add("d-none");
  }
 }

 searchInput.addEventListener("input", filterProducts);
 categoryRadios.forEach((radio) => radio.addEventListener("change", filterProducts));

 // 3. Price Sorting via Existing DOM Node Reordering (appendChild)
 sortBtn.addEventListener("click", () => {
  if (sortDirection === "none" || sortDirection === "desc") {
    sortDirection = "asc";
    sortBtn.innerHTML = '<i class="bi bi-sort-numeric-down me-1"></i> Price: Low to
High';
    sortBtn.classList.replace("btn-outline-dark", "btn-dark");
  } else {
    sortDirection = "desc";
    sortBtn.innerHTML = '<i class="bi bi-sort-numeric-down-alt me-1"></i> Price: High to
Low';
    sortBtn.classList.replace("btn-dark", "btn-outline-dark");
```

```
    }

    const sortedCards = [...productCards].sort((a, b) => {
      const priceA = parseFloat(a.dataset.price) || 0;
      const priceB = parseFloat(b.dataset.price) || 0;
      return sortDirection === "asc" ? priceA - priceB : priceB - priceA;
    });

    sortedCards.forEach((card) => productGrid.appendChild(card));
  });
});
```